Term Project Report

Vehicle Suspension Dynamics

MCAE 210 – Dynamics

Instructor: **Prof. Bahman Naser**

Author: **Ivan Bednostin**

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School of Engineering

Department of Civil & Mechanical Engineering

Shippensburg University of Pennsylvania

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ABSTRACT

The main purpose of this experiment is to simplify a real-world vehicle suspension system into a one-degree-of-freedom dynamic model and derive its governing ordinary differential equation using Newton's laws and fundamental principles of mechanics. The study then solves for the displacement response $x(t)$ under sinusoidal base (road) excitation and obtains a particular solution using the given system parameters. The resulting model, based on this particular solution, is used to analyze the dynamic behavior of the specific vehicle configuration and to determine how the response amplitude, oscillation severity, and rate of decay depend on the spring constant k and damping coefficient c.

The analytical part of the project produced a governing second-order differential equation describing the motion of the vehicle suspension system. The model includes the mass of the vehicle (inertia effects), spring restoring force, damping force, and sinusoidal base excitation from the road. Solving this equation resulted in the displacement response $x(t)$ for the specified system parameters.

The second set of results consists of simulation graphs and animations obtained in MATLAB and Simulink, which provide a visual representation of the system response. These results allow direct comparison between the baseline case (given parameters) and modified parameter values. The graphs clearly show how changes in the spring constant and damping coefficient affect the amplitude of displacement and the overall vibration behavior of the system.

Numerical simulations in MATLAB and Simulink validate the analytical results, illustrating the effects of varying spring stiffness and damping on displacement amplitude, phase, and transient decay. Low damping produces prolonged oscillations, whereas increased damping accelerates convergence to steady-state motion. Resonance occurs when excitation frequency approaches the natural frequency, amplifying displacement, emphasizing the importance of proper parameter selection for comfort and stability. These combined analyses provide a comprehensive understanding of the suspension dynamics and the interplay of mass, stiffness, and damping in controlling vehicle response.

INTRODUCTION

Real-world mechanical systems are often too complex to analyze directly, as they involve multiple interacting components and degrees of freedom. A vehicle suspension system, in particular, includes several masses, rotational effects, and interactions between multiple elements. To make analysis possible, such a system can be reduced to a simplified one-degree-of-freedom model consisting of a mass, a spring, and a damper, which represents the primary vertical motion of the vehicle body.

This simplification neglects effects such as wheel inertia and rotational dynamics of the vehicle body about its center of mass (e.g., pitch), which in real conditions leads to redistribution of normal forces between the front and rear wheels. Despite these omissions, the simplified model retains the dominant forces acting on the system: inertia, spring

restoring force, and damping force. This makes it possible to study how these forces interact and how their governing parameters influence the motion.

In addition, the road profile is also simplified and modeled as a sinusoidal base excitation. While real road conditions are irregular and non-periodic, this idealized input provides a controlled and consistent case for analysis. Together, these simplifications allow the system to be described by a single differential equation, making it possible to isolate and clearly observe the relationships between system parameters and the resulting vibration response.

OBJECTIVE

The objective of this experiment is to formulate a one-degree-of-freedom model of a vehicle suspension system and derive its governing equation of motion using Newton's Second Law. The model is then used to determine the displacement response $x(t)$ under sinusoidal base excitation for the given system parameters. In addition, the objective is to evaluate how variations in the spring constant k and damping coefficient c influence the system response.

THEORY

The motion of the suspension system is governed by Newton's Second Law, which states that the net force acting on the vehicle body is equal to the mass times its acceleration. To describe the forces acting on the system, the spring force is proportional to the relative displacement between the vehicle body and the road input, while the damping force is proportional to the relative velocity. Let $x(t)$ be the absolute displacement of the vehicle body and $y(t)$ the road displacement. Then,

$$F_s = -k(x - y), \qquad F_d = -c(\dot{x} - \dot{y})$$

where k is the spring constant and c is the damping coefficient.

Applying Newton's Second Law to the mass, the governing equation of motion becomes:

$$m\ddot{x} + c(\dot{x} - \dot{y}) + k(x - y) = 0$$

Here, m is the vehicle mass, and the motion is measured from the static equilibrium position. Therefore, the weight (mg) of the vehicle does not appear explicitly in the final equation, since it is already balanced by the static deflection of the spring.

Next, the external excitation must be defined. For a sinusoidal road input,

$$y(t) = Asin(\omega t)$$

the road acts as a base excitation.

To understand how the system responds to this input, the concept of natural frequency must be introduced, since the response is governed by the relationship between the excitation frequency (road input) and the system's natural frequency. The natural frequency is the frequency at which the system oscillates on its own after being disturbed and then released without any external forcing. It is called natural because it depends only on the system properties, namely the mass and the spring stiffness.

For example, if you apply slight pressure to a car and then release it, it will begin to oscillate up and down. The frequency of these oscillations represents the natural frequency of the suspension. Similarly, a weight attached to a spring, when stretched and subsequently released, will oscillate at its own inherent characteristic frequency. For a given system, the natural frequency is determined as follows:

$$\omega_n = \sqrt{\frac{k}{m}}$$

and in hertz it is

$$f_n = \frac{\omega_n}{2\pi}$$

The importance of this quantity becomes clear when comparing it with the excitation frequency. When the excitation frequency approaches the natural frequency, the system enters a resonance condition, where the displacement amplitude increases significantly. Physically, this happens because the external input adds energy to the system in phase with its motion, meaning each cycle reinforces the oscillation instead of damping it out.

In practical terms, if the road excites the suspension at a frequency close to its natural frequency, even small bumps can produce large vibrations of the vehicle body.

Finally, it is necessary to consider the role of damping. The critical damping ratio is defined as:

$$c_{cr} = 2\sqrt{km}$$

This value represents the boundary between oscillatory and non-oscillatory system behavior. In the event of a disturbance—for instance, striking a single bump followed by a return to a level road—critical damping ensures the most rapid return to a state of equilibrium without the onset of oscillations. When the damping falls below this value, the system exhibits oscillatory behavior. Conversely, if the damping exceeds the critical value, the system becomes overdamped and returns to equilibrium more slowly, although oscillations are completely suppressed in the process.

If the specified damping constant of the given damper was divided by the value of critical damping, the damping ratio would be obtained. Essentially, the implications of this concept align with the idea that was presented above; the resulting value is now simply compared to 1 and subsequently used to determine the solution for $x(t)$:

$$\zeta = \frac{c}{c_{cr}} = \frac{c}{2\sqrt{km}}$$

Here are some qualitative responses of the system:

- $\zeta<1$: underdamped (oscillatory response with decay)
- $\zeta=1$: critically damped (fastest return without oscillations)
- $\zeta>1$: overdamped (slow, non-oscillatory response)

Physically, damping represents energy dissipation. A small damping ratio implies that very little energy is removed from the system, allowing oscillations to persist over time. In this study, the damping ratio is extremely small ($\zeta \ll 1$), indicating a highly underdamped system.

Up to this point, the governing equation has been developed and analyzed in its analytical second-order form. However, for practical evaluation of the system response under different parameter values and input conditions, a numerical solution approach is required.

Most computational solvers operate on systems of first-order differential equations rather than directly solving second-order equations. In this experiment, MATLAB was used as the primary computational tool, which also requires the system to be expressed in first-order form for numerical integration.

To achieve this, the following state variables are introduced:

$$x_1 = x, \qquad x_2 = \dot{x}$$

This allows the original equation to be rewritten as a system:

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = \frac{1}{m}\left[-\, c(x_2 - \dot{y}) - k(x_1 - y)\right]$$

where the road input is defined as

$$y(t) = A\sin(\omega t), \qquad \dot{y}(t) = A\omega\cos(\omega t)$$

This transformation does not change the physical meaning of the model, but expresses it in a form suitable for numerical simulation. As a result, the system can be implemented in MATLAB to evaluate the time-dependent response $x(t)$ under various conditions.

APPARATUS AND INSTRUMENTATION

This project was conducted using analytical methods and computational tools rather than physical laboratory equipment. The primary software used was MATLAB and Simulink (MathWorks Inc.), each serving a distinct purpose in the analysis.

The MATLAB environment was utilized to calculate and visualize the time-dependent system response to a sinusoidal road disturbance. The generated plots displayed both the input excitation (road profile) and the resulting displacement of the vehicle body over extended time intervals, thereby enabling observation of steady-state behavior and resonant effects. In addition, the implemented code allowed for enhanced visualization through synchronized animations of the system response, providing a more intuitive representation of the interaction between the road input and the motion of the suspension.

Simulink was used to model the system response to a single-bump disturbance, representing a more realistic road condition. This approach allowed direct observation of the transient response, including how quickly oscillations decay and how the system returns to equilibrium under different values of the spring constant k and damping coefficient c.

TEST PROCEDURE

The procedure of this study consisted of both analytical derivation and numerical simulation in order to evaluate the dynamic response of a simplified suspension system.
First, the governing equation of motion was derived using Newton's Second Law by accounting for inertia, spring force, damping force, and base excitation from the road. This stage enabled the formulation of the mathematical model required for all subsequent analytical calculations.

Next, the equation was expressed in a form suitable for numerical computation and implemented in MATLAB. The system response was evaluated over a specified time interval under sinusoidal road excitation. This step was essential for observing steady-state behavior and identifying conditions under which resonance occurs.

Furthermore, the spring stiffness constant k and the damping coefficient c were systematically varied, while all other parameters (such as the vehicle mass and the nature of the input disturbance) remained constant. This approach allowed for an isolated assessment of the influence of each of these parameters on the amplitude and decay characteristics of the oscillations.

To conduct a more detailed investigation of transient processes, a separate model was created in Simulink to simulate the system's response to a single road obstacle (specifically, a pothole). The system's response was analyzed across various damping coefficient values to determine the rate of oscillation decay and the manner in which the system returns to its equilibrium state.

Throughout the procedure, the mass of the system and the general form of excitation were kept constant to ensure consistent comparison between different test cases. The independent variables were the spring constant k and damping coefficient c, as these parameters directly control stiffness and energy dissipation in the system. These variables were selected because they are the primary design parameters in suspension systems and have the most significant influence on vibration amplitude, resonance behavior, and ride comfort.

RESULTS AND DISCUSSION

Analytical Solution and Component Interpretation

Overall, as briefly mentioned earlier, the solution to the second-order linear differential equation describing a mass-spring-damper system consists of homogeneous and particular components. For underdamped systems ($c < 2\sqrt{km}$), the response is oscillatory in nature, with an exponentially decaying amplitude, and the displacement can be expressed as the sum of two physically distinct components: a transient (homogeneous) part and a steady-state (forced) part:

$$x(t) = e^{-\zeta\omega_n t}(C_1 cos(\omega_d t) + C_2 sin(\omega_d t)) + Bsin(\omega t) + Ccos(\omega t)$$

The homogeneous component represents the natural dynamics of the system and follows an exponential–oscillatory form governed by the natural frequency $\omega_n = \sqrt{k/m}$ and the damped frequency $\omega_d = \sqrt{1 - \zeta^2}$. The particular component represents the steady-state response to the external sinusoidal excitation and oscillates at the excitation frequency.

This general representation is characteristic of all underdamped second-order systems and provides a direct framework for constructing the full solution. By substituting the given system parameters into this form and solving for the constants C_1 and C_2, the complete response is obtained. In this case, no independent initial displacement $x(0)$ or velocity $\dot{x}(0)$ is imposed; instead, the motion is entirely driven by the external excitation. As a result, the constants are determined consistently with the forced input conditions rather than arbitrary initial states.

The resulting displacement function is:

$$x(t) = e^{-0.015t}(0.000338cos(18.708t) + 0.15032sin(18.708t)) - 0.08054sin(34.9t) - 0.000338cos(34.9t)$$

Only the standard relationships for underdamped systems were required to construct this solution, without introducing additional transformations into a canonical form. The full derivation, including substitution of parameters and determination of constants, is provided in Appendix A.

Having established the analytical form of the solution, the physical interpretation of its components can now be examined. Transient effects correspond to the homogeneous solution of the governing differential equation and represent exponentially decaying oscillations determined solely by system parameters. In contrast, the steady-state component corresponds to the particular solution and reflects the direct response to external forcing.

Since the excitation is introduced as a sinusoidal road input from the initial moment, both components are present simultaneously from $t = 0$, making them difficult to distinguish at early times despite their different physical origins. As a result, the initial response appears as a combination of both behaviors.

For low damping values, the response initially deviates from a purely sinusoidal form. Specifically, this deviation appears as slight waveform distortion and variation in amplitude, while the motion remains predominantly oscillatory. In addition, a noticeable phase lag relative to the road input is observed, indicating that the system does not respond instantaneously due to inertia and stiffness effects.

This behavior is clearly illustrated in the Desmos visualizations (see Fig. 1).

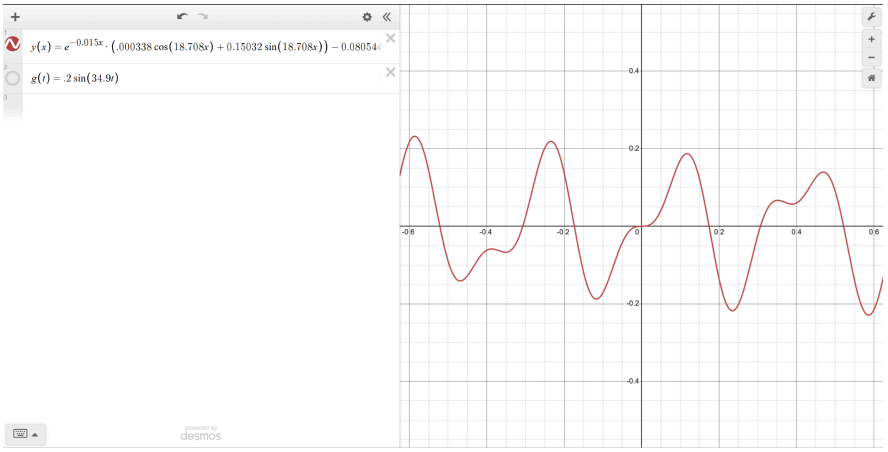


Fig. 1. Early-time response of the system showing distorted oscillations and visible influence of transient effects.

At early times, the curve appears irregular rather than perfectly sinusoidal, although it remains bounded within expected amplitude limits. This reflects the superposition of transient and steady-state components.

To further investigate this behavior, the solution can be examined over an extended time interval (see Fig. 2).

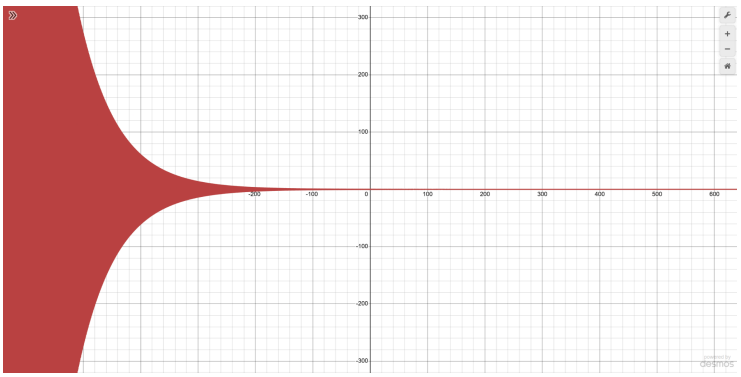


Fig. 2. Extended-time view (including negative time) showing oscillations with exponentially varying amplitude.

In the negative time direction, the amplitude increases exponentially, which is consistent with the mathematical form of the homogeneous solution. This observation confirms that the transient component is inherently present but decays in forward time due to damping.

As time progresses, the transient contribution gradually diminishes (see Fig. 3).

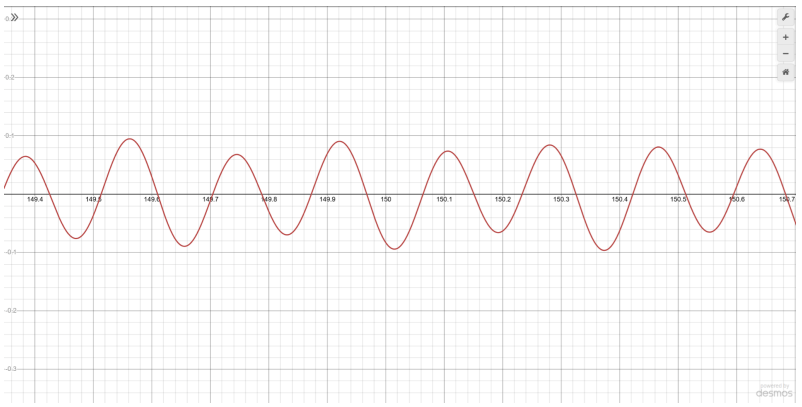


Fig. 3. Intermediate-time response illustrating decay of transient effects and reduction of amplitude variations.

After ~100 seconds, the waveform becomes smoother, and differences between successive amplitudes decrease. This indicates that the transient component is being effectively suppressed by damping.

Eventually, the response approaches steady-state behavior (see Fig. 4).

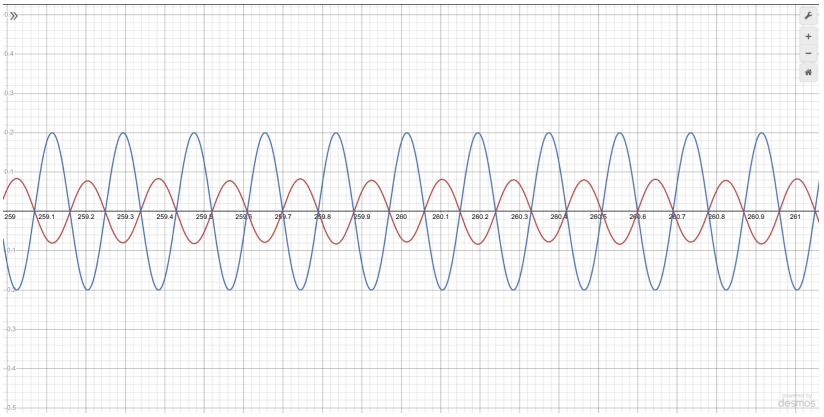


Fig. 4. Long-term response showing a nearly sinusoidal steady-state motion dominated by external excitation.

Beyond ~250 seconds, the motion closely resembles a pure sinusoid. At this stage, the influence of the homogeneous component becomes negligible, and the system response is governed almost entirely by the external forcing. The amplitude stabilizes at a reduced level (approximately 0.1 in this case), and the waveform aligns with the excitation frequency.

Furthermore, in this mode, a constant phase shift is observed between the input excitation and the response. The phase lag is determined by the relationship between the excitation frequency, the natural frequency, and the damping. The observed shift approaches half a period (π), indicating operation in a regime where the excitation frequency is relatively high compared to the system's natural frequency.

Overall, the results demonstrate that damping plays a critical role in suppressing transient effects, allowing the system to transition from a mixed response to a purely forced oscillation governed by the road input.

MATLAB Simulations: Baseline and Parameter Variation

To gain a more detailed understanding of the system's behavior, numerical simulations were conducted using MATLAB and Simulink. These tools are particularly valuable because the analytical dependence of displacement, decay rate, amplitude variation, and phase shift on given parameters is often difficult to interpret solely by relying on formulas. Furthermore, generating new plots in Desmos requires significant time and effort. In contrast, graphical results allow for the rapid investigation of interdependencies between parameters and facilitate their clear visual comparison across various sets of values.

Using MATLAB, it is possible to plot the sinusoidal road profile simultaneously with the vehicle's displacement graph. The software automatically substitutes all necessary parameters and recalculates function values at every step, thereby enabling experimentation with a wide range of scenarios—including extreme or unconventional cases. In this sense, MATLAB and Simulink serve as a practical complement to the analytical solution, offering the ability to quickly and visually verify the impact of parameter variations.

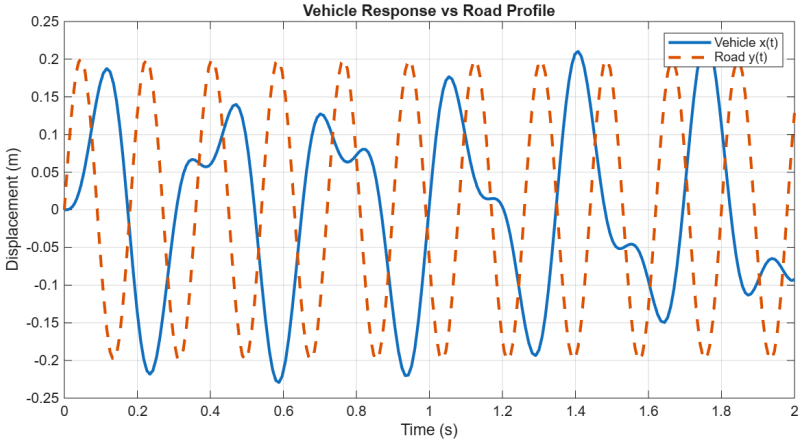
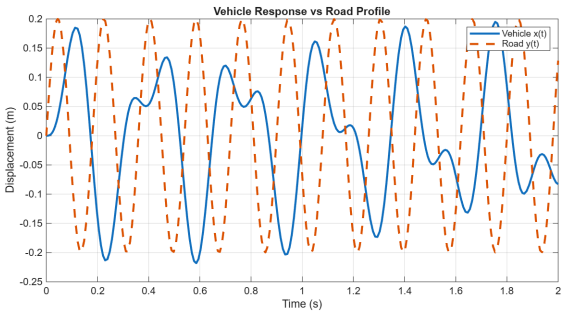


Fig. 5. Plot of the road surface profile and vehicle displacement, generated in MATLAB for the baseline scenario.

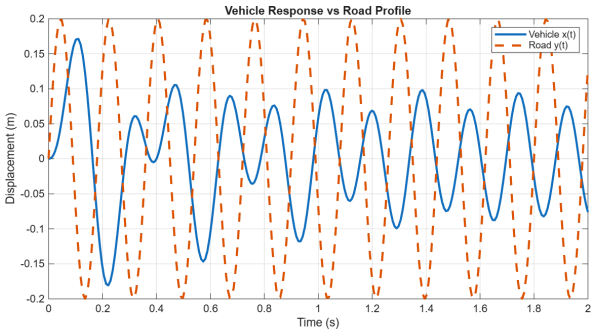
Note: The baseline scenario assumes that all other parameters remain unchanged as specified in the problem statement, while the experiment is conducted by varying the selected parameter (damping or spring stiffness).

The results of the MATLAB simulations confirm the general system dynamics predicted analytically, while providing additional insight into the role of damping and spring stiffness in shaping the displacement response. When the displacement graph is superimposed onto the road surface profile, the behavior of the vehicle body under different parameter combinations becomes evident.

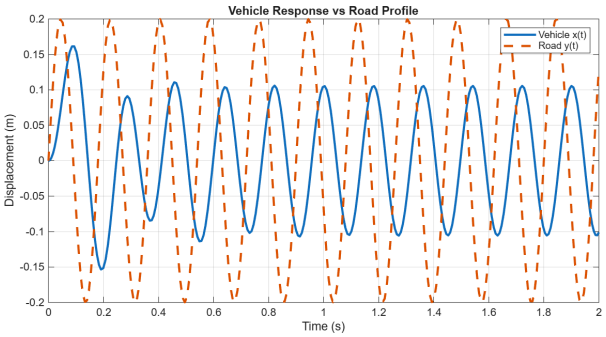
For very low damping values (e.g., 30), MATLAB does not effectively resolve the decay of initial oscillations, and the displacement remains nearly unchanged, failing to show a transition toward a steady-state sinusoidal response as is visible in the analytical solution and in Desmos. However, for moderate and higher damping values (e.g., 3000 and 10000), the initial transient oscillations are quickly suppressed by the damping force, and the system approaches a sinusoidal motion at the excitation frequency. Although the resulting phase shift differs from that observed in Desmos, a clear transition is now visible: the early-time natural oscillations are damped, and a stable sinusoidal pattern emerges (see Fig. 6a–c).



a.



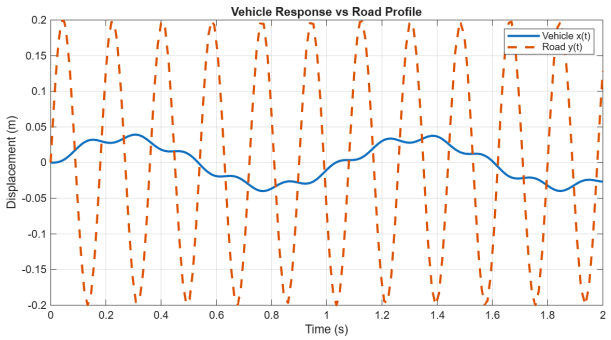
b.



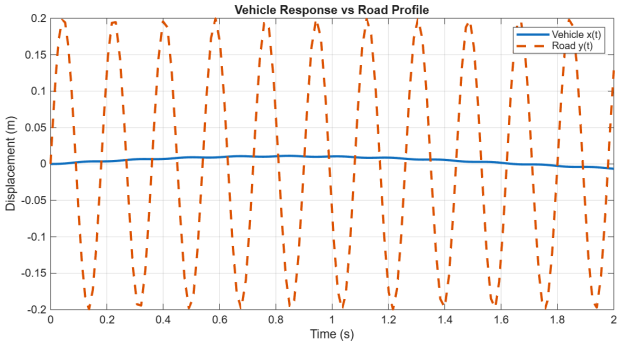
c.

Fig. 6a–c. Comparison of displacement dynamics for various damping values (300, 3000, and 10000).

The influence of spring stiffness is also evident. Reducing the spring constant (e.g., from 35000 to 1000) drastically decreases the amplitude of the oscillations, particularly when damping is small. In this case, the weak spring cannot transmit sufficient restoring force to the vehicle body, resulting in almost negligible displacement during the bump input. Introducing a damping value close to the critical level restores the oscillatory amplitude, producing a smooth sinusoidal response without pronounced transitional oscillations. This combination—a lower spring constant with critical damping—yields a more comfortable response, as the vehicle body follows the road profile smoothly from the start, avoiding excessive initial transients (see Fig. 7a–c).



a.



b.

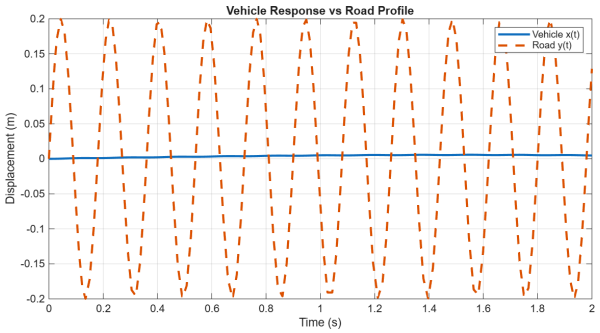


Fig. 7a–c. Comparison of displacement dynamics for various spring stiffness values (35000, 3500, and 1000).

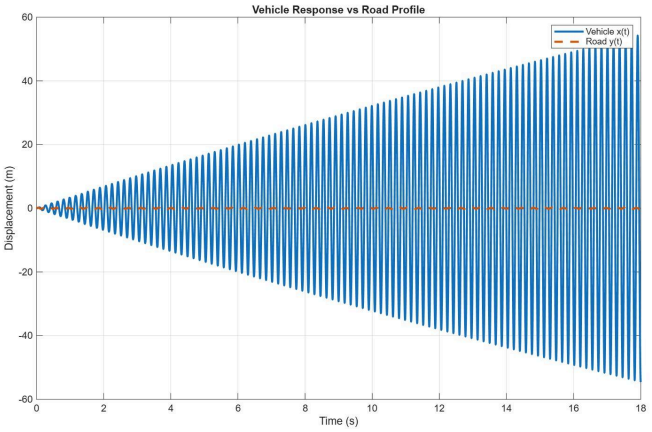
These figures illustrate how damping controls the suppression of transient oscillations, while spring stiffness governs the magnitude of the displacement. Together, they determine not only the amplitude but also the quality of the vehicle's response to road irregularities, and highlight that appropriate tuning of both parameters is critical for achieving smooth, comfortable motion.

Resonance Behavior and the Effect of Damping

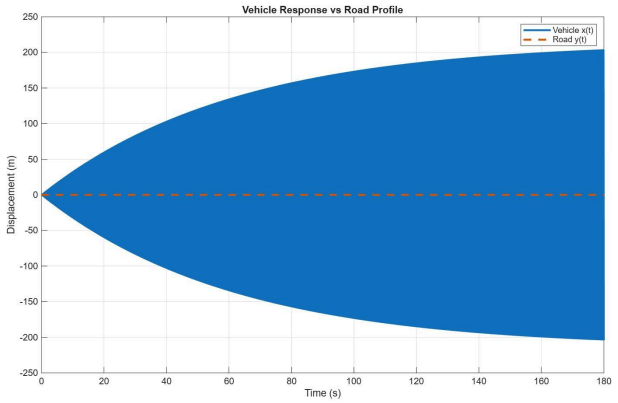
MATLAB simulations provide a clear demonstration of resonance phenomena and the critical role of damping in controlling system response. Resonance occurs when the excitation frequency approaches the system's natural frequency, $\omega_n = \sqrt{k/m}$. Under these conditions, even small external forces can produce large amplitude oscillations.

To illustrate the phenomenon, simulations were conducted over 100 and 1000 excitation periods. For low damping values, the amplitude initially grows rapidly, reaching significant levels before gradually converging toward a steady oscillatory state. Increasing the damping coefficient reduces both the peak amplitude and the time required to approach this steady state. These cases provide a visually intuitive understanding of how resonance develops and stabilizes over time (see Fig. 8a–b).

All additional simulations—including longer durations (6000 periods) and varied damping values—are provided in Appendix B for reference. These extended results reinforce the same trends: damping strongly moderates amplitude growth and determines the final oscillatory behavior, even under extreme resonance conditions.



a.



b.

Fig. 8a–b. Resonance response for 100 and 1000 periods under varying damping values: initial amplitude growth and subsequent convergence.

From an engineering perspective, understanding resonance is critical. Large oscillations at or near the natural frequency can pose safety risks, lead to material fatigue, or compromise vehicle stability and comfort. The simulations illustrate how appropriate damping selection mitigates these risks, preventing excessive displacement and ensuring that the system responds in a controlled and predictable manner. Moreover, visualization of resonance in

MATLAB provides an effective tool for exploring the interaction between damping and stiffness, and for designing systems that avoid dangerous resonance conditions.

Response to a Single Pothole in Simulink: Damping and Elasticity Effects

The influence of these parameters is particularly evident when traversing road irregularities, such as potholes. Simulink is utilized to model this scenario: the system generates a model of a pothole defined by specific, precise parameters—spanning from the crest of a cosine profile down to its lowest point—after which the road continues as a level section. This approach enables rapid modification of parameters and the instantaneous generation of new results, thereby facilitating the swift evaluation of numerous design variations without incurring additional effort.

In instances of insufficient damping—as observed under the initial conditions—the inertial mass continues to oscillate after clearing the pothole, resulting in a prolonged transient response. This phenomenon stems from the fact that high spring stiffness amplifies the system's reaction: even minor displacements of the vehicle body generate a significant restoring force that sustains these oscillations. Conversely, in cases where the damping level approaches the critical threshold, the vehicle body returns to a state of equilibrium significantly faster, regardless of the specific spring stiffness coefficient.

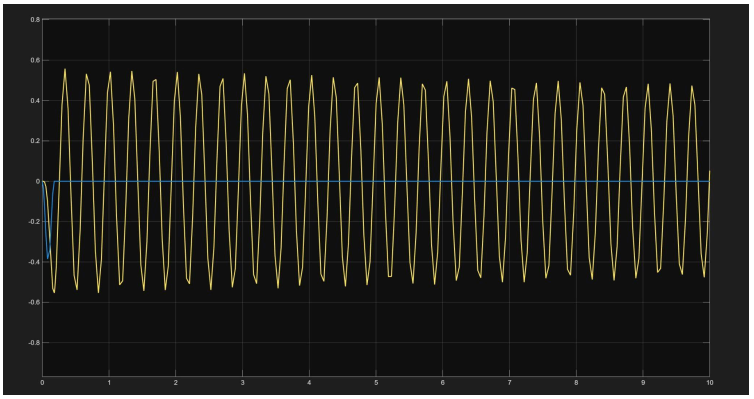


Fig. 8. Vehicle body dynamics when traversing a pothole, with low damping and original spring stiffness.

Note: Observe how oscillations persist long after the vehicle has passed over the irregularity.



Fig. 9. Vehicle body dynamics with critical damping and the same spring stiffness.

Note: Oscillations are damped quite rapidly, and the return to equilibrium occurs almost instantaneously.

Additionally, a series of experiments was conducted involving variations in the spring stiffness coefficient. The spring stiffness values were progressively reduced, and the optimal critical damping level was determined individually for each case. As stiffness is reduced, the following effects are observed: the initial downward displacement of the vehicle body decreases, as a softer spring transmits less force; the subsequent upward rebound of the spring also exerts a less intense impact on the body, and the amplitude of subsequent oscillations is noticeably reduced.

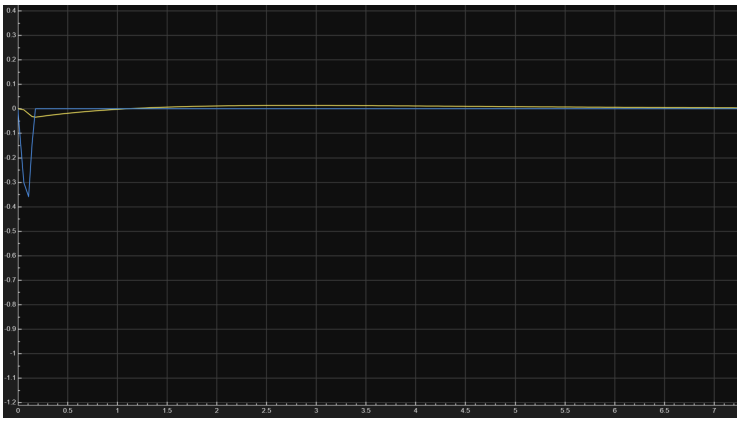


Fig. 10. The effect of reduced spring stiffness ($350\ \frac{N}{m}$) on the vehicle body's return to equilibrium under critical damping ($1183.22\ \frac{kg}{s}$).

Note: The vehicle body returns slowly to its original position, remaining slightly elevated for a longer duration.

A complete set of additional comparisons for various combinations of stiffness and damping is provided in Appendix C.

CONCLUSION

The present study investigated the dynamic behavior of a mass-spring-damper system subjected to sinusoidal road forces and isolated road irregularities (potholes) using both analytical methods (Desmos) and numerical simulations (MATLAB and Simulink). The goal was to explore the fundamental principles of vibration, damping, and stiffness control, and to demonstrate how simplified single-degree-of-freedom models can capture the essential characteristics of more complex real-world systems, thereby reducing computational and conceptual complexity without sacrificing understanding. The main findings of this investigation are summarized below:

1. *Analytical and numerical consistency*. The analytical solution, decomposed into transient (homogeneous) and steady-state (forced) components, provides an accurate basis for understanding the behavior of the system. Transient effects appear as exponentially decaying oscillations determined solely by system parameters, while the

steady-state response reflects the continuous influence of the external sinusoidal input signal. Desmos visualizations clearly show transient decay and subsequent establishment of sinusoidal steady motion, supporting the analytical framework. MATLAB simulations capturing the system response reveal a noticeable difference: for very low damping values, the transition from initial oscillations to a steady-state sinusoidal response is not visualized; the displacement graph appears “stable” from the first periods, highlighting differences in numerical resolution between software tools.

2. *Effect of damping on transient and steady-state response*. Damping plays a crucial role in controlling the decay of transient oscillations. For low damping values, transient effects persist for longer periods, producing pronounced deviations from the sinusoidal input and extended transitional behavior. As damping increases, the transient oscillations are rapidly suppressed, allowing the system to achieve steady-state motion more quickly. Critically, higher damping reduces the amplitude of overshoot during transient motion and stabilizes the response, while low damping prolongs the dominance of natural oscillations, delaying convergence to the steady-state profile.
3. *Influence of spring stiffness*. Spring stiffness primarily governs the magnitude of displacement and the natural frequency of oscillation. Higher stiffness results in larger restoring forces, faster oscillatory motion, and increased sensitivity to external disturbances. Conversely, reducing stiffness diminishes the initial displacement amplitude, particularly in combination with low damping, leading to smaller oscillations but a slower return to equilibrium, as well. Experiments show that optimal damping combined with appropriately selected stiffness can provide a smooth and comfortable ride, minimizing transient vibrations while maintaining sufficient dynamic responsiveness.
4. *Response to road irregularities* (single pothole). Simulations in Simulink allow detailed exploration of single-event disturbances. Low damping values, combined with strong springs, produce prolonged oscillations after a bump, illustrating the inertia-driven overshoot and extended transient behavior. When damping is near the critical value, the system rapidly returns to equilibrium, even for varying spring constants. Decreasing spring stiffness reduces the amplitude of initial displacement, while critical damping ensures a smooth, monotonic return to equilibrium. These results highlight the interplay between damping and stiffness in mitigating uncomfortable vehicle responses to road irregularities.
5. *Resonance phenomena*. The study confirms that resonance occurs when the excitation frequency approaches the system’s natural frequency. MATLAB simulations demonstrate that under low damping conditions, resonance produces rapid amplitude growth over multiple periods before stabilizing at a steady-state amplitude. Increasing damping attenuates the peak amplitude and shortens the time to reach steady-state oscillations. This observation underscores the engineering importance of avoiding resonant conditions, as excessive amplitudes can compromise safety, structural integrity, and passenger comfort. Visualizing resonance also provides an intuitive understanding of how damping and stiffness interact to control amplitude under extreme dynamic conditions.
6. *Practical insights and reduced complexity modeling*. The combination of analytical and numerical methods shows how single-degree-of-freedom models capture the important dynamics of more complex systems, offering engineers a flexible approach to predicting and optimizing system behavior. By reducing complexity while maintaining important characteristics, such models allow systematic exploration of

parameter spaces, including damping, stiffness, and excitation frequency, without excessive computational effort. Numerical tools, such as MATLAB and Simulink, complement analytical solutions by enabling rapid scenario testing, immediate visualization of dynamic response, and iterative parameter tuning to achieve desired results.

To summarize, this study highlights that efficient vehicle suspension design depends on careful tuning of damping and stiffness parameters. Proper damping quickly suppresses transients and prevents excessive oscillation, while spring rate determines the amount of displacement and dynamic response. The insights gained from single-degree-of-freedom modeling, supported by both analytical and numerical approaches, provide a strong foundation for understanding complex real-world systems in a simplified but strict model. The methodology demonstrated here illustrates a practical way in which engineers can predict, visualize, and optimize dynamic behavior while reducing model complexity, which is critical in both research and industrial applications.

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APPENDICES

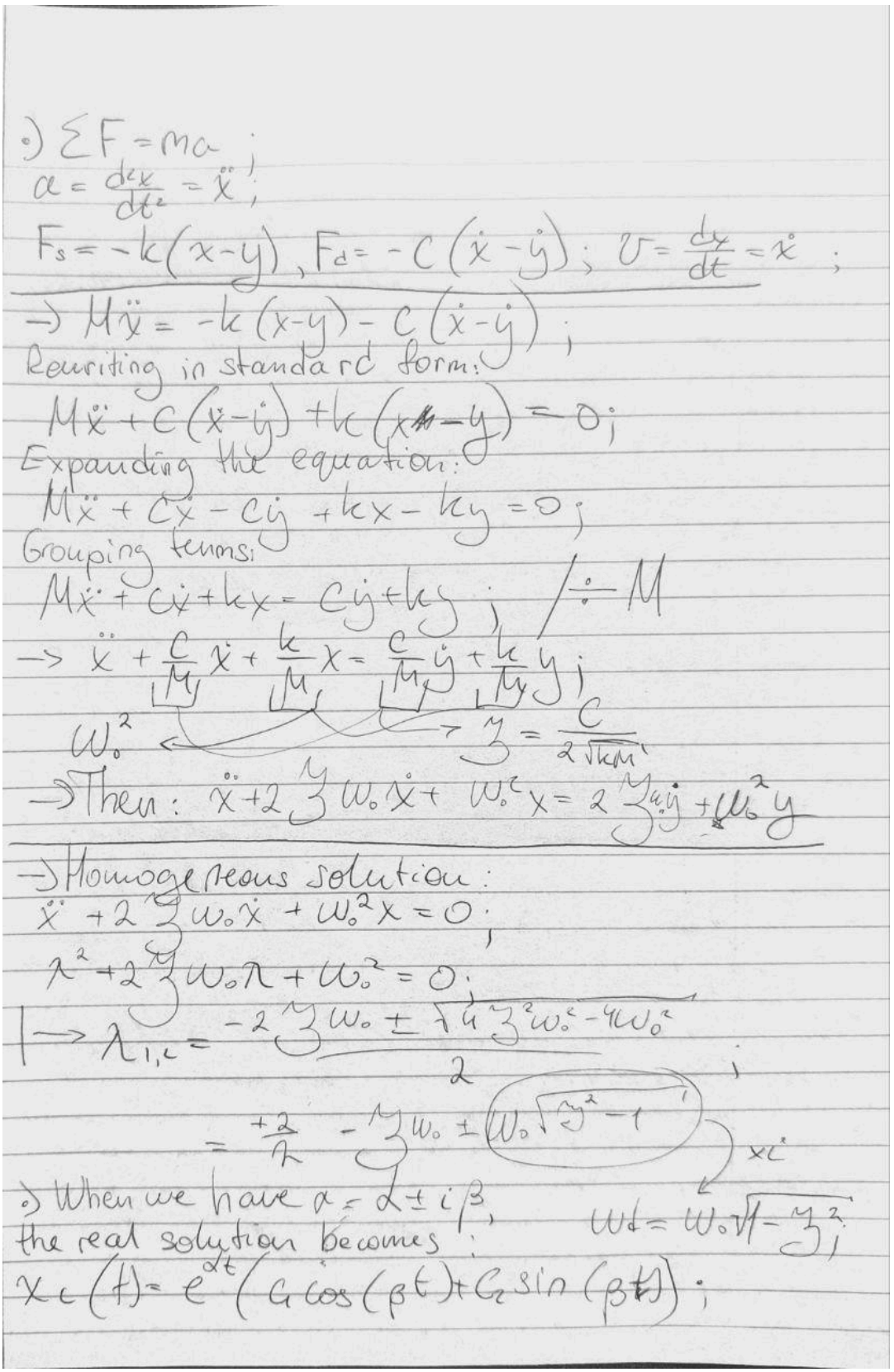


Figure A1. Newton's second law formulation, Derivation of the differential equation, Homogeneous solution of the ODE.

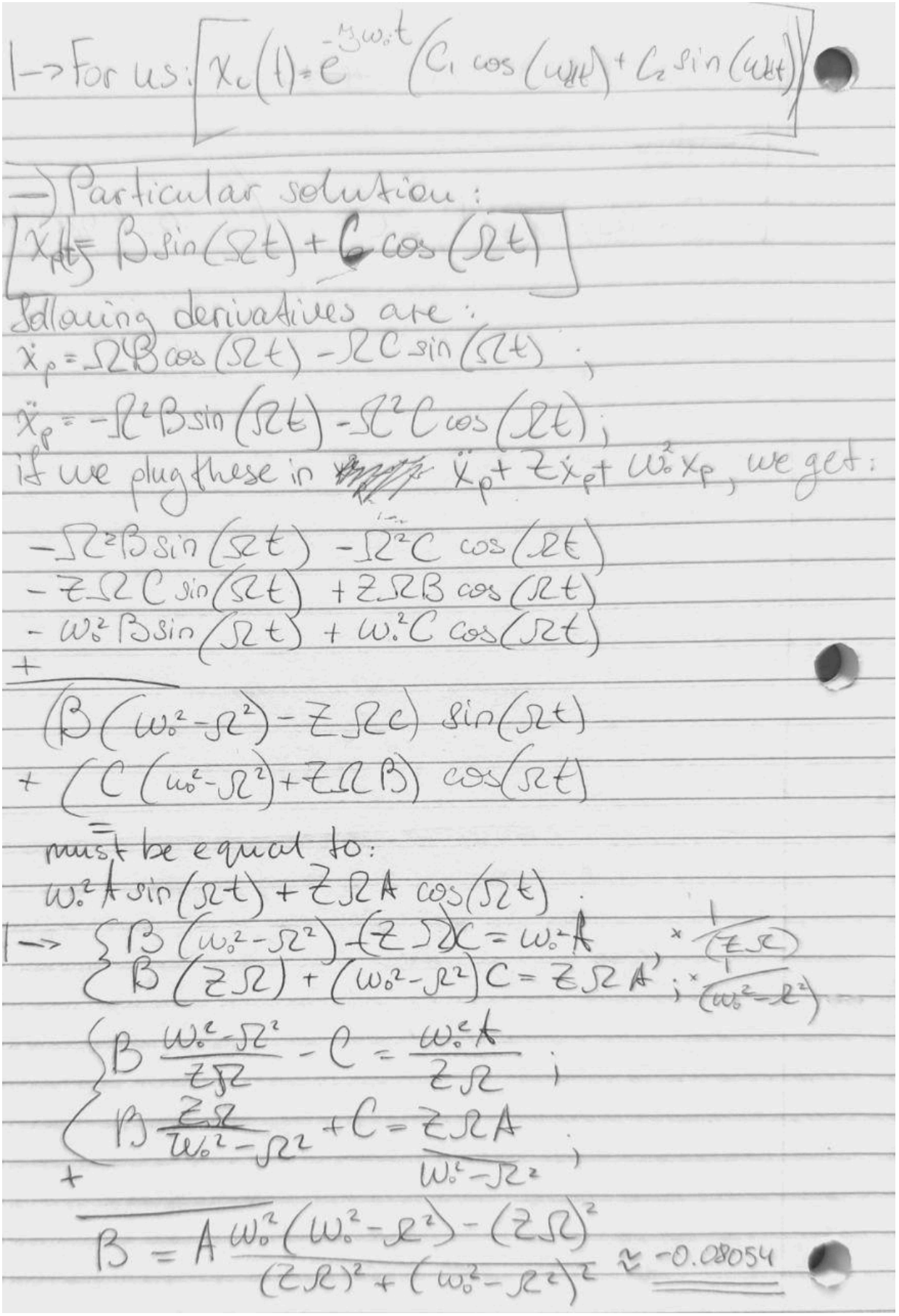


Figure A2. Particular solutions of the ODE (solving for B coefficient).

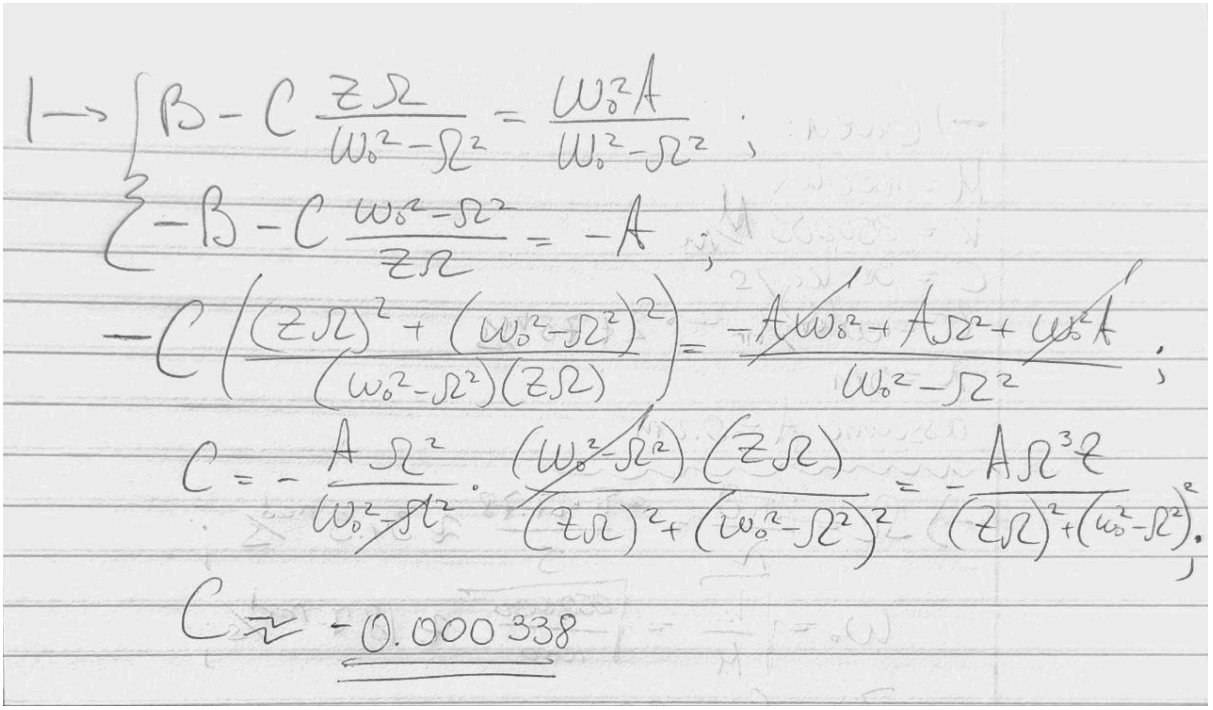


Figure A3. Particular solutions of the ODE (solving for *c* coefficient).

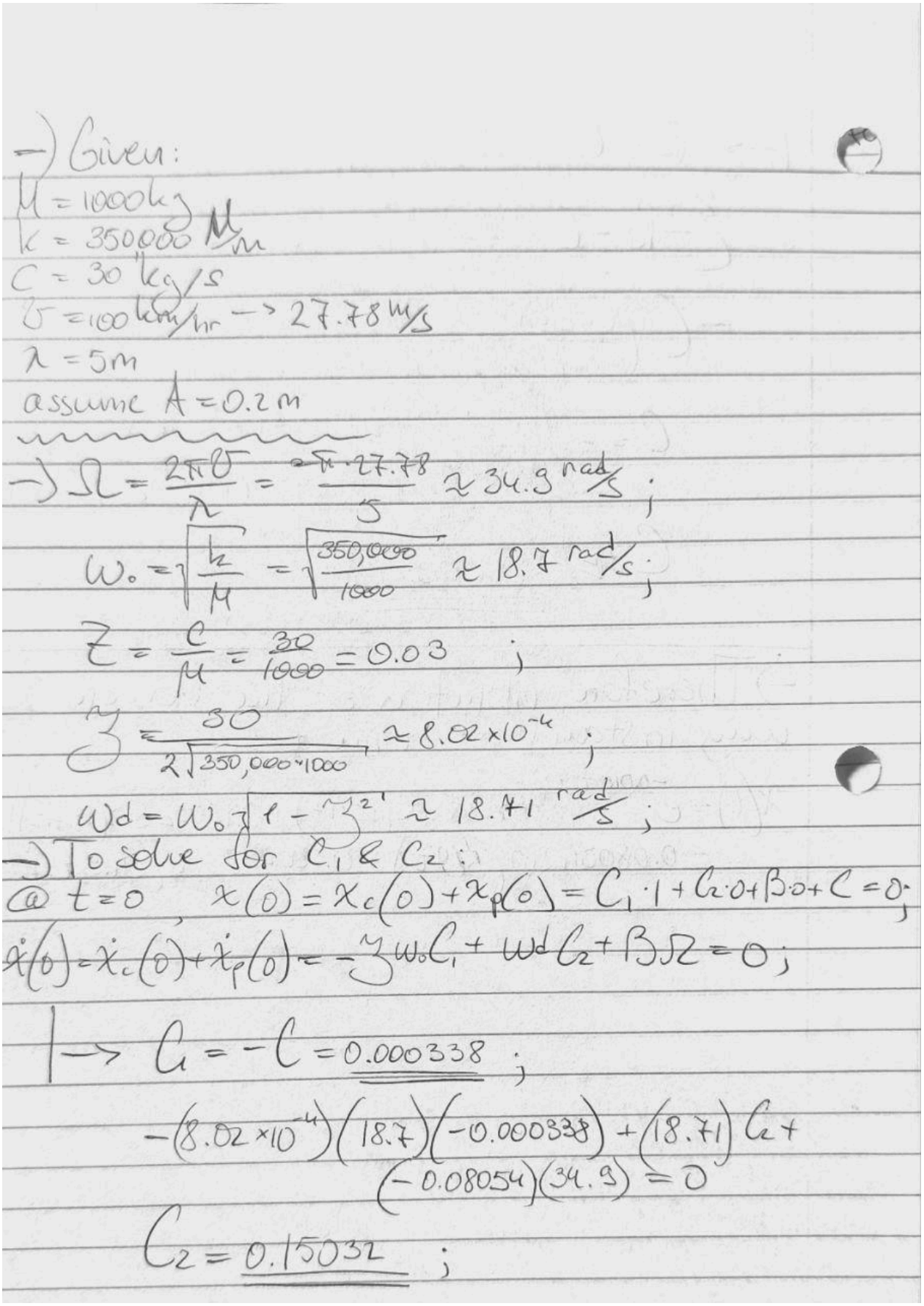


Figure A4. Determination of constants in the particular solution, Substitution of given parameters.

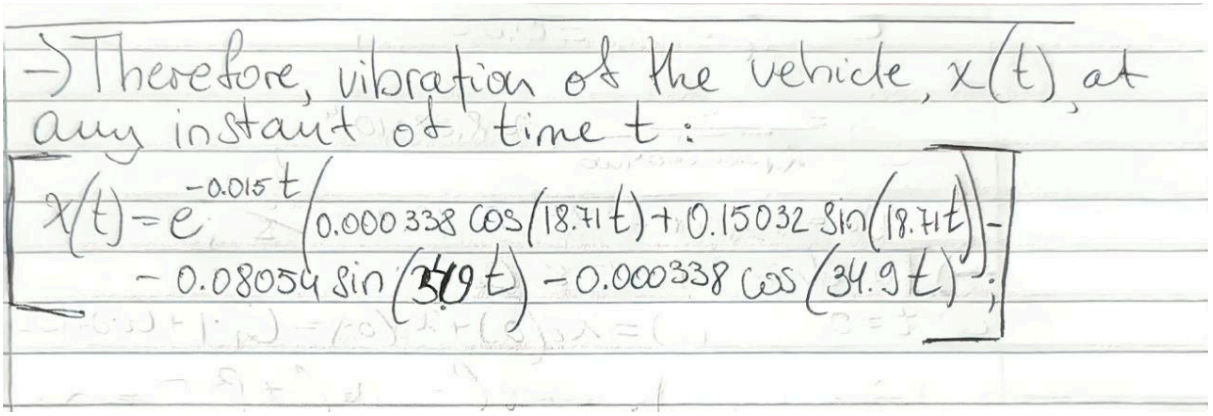


Figure A5. Final displacement function.

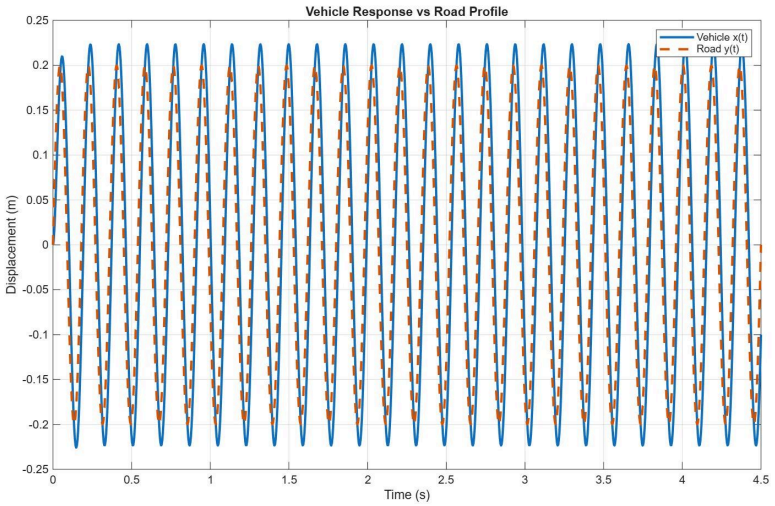


Figure B1. Resonance simulation of ~25 periods with critical damping.

Displacement closely follows the sinusoidal road input, with slight amplitude increase over the excitation profile.

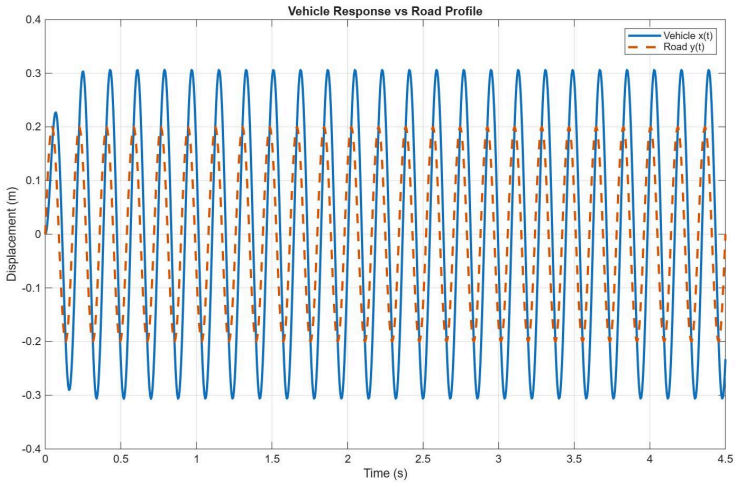


Figure B2. Resonance simulation of ~25 periods with damping $c = 30000$.

Displacement reaches $\sim$ 30% *higher amplitude than the road input, while still maintaining a stable sinusoidal pattern.*

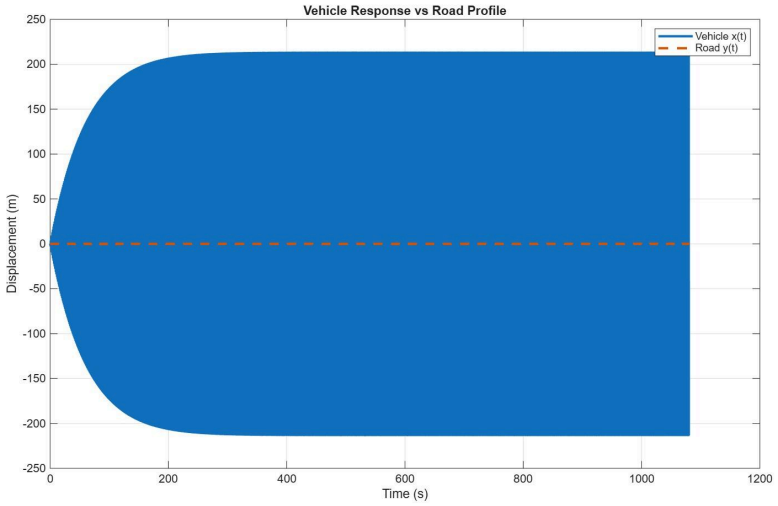


Figure B3. Resonance simulation over 6000 periods with baseline damping.

The displacement graph shows rapid amplitude growth initially, then stabilizes at a consistent level, appearing nearly as a continuous block due to the high number of periods.

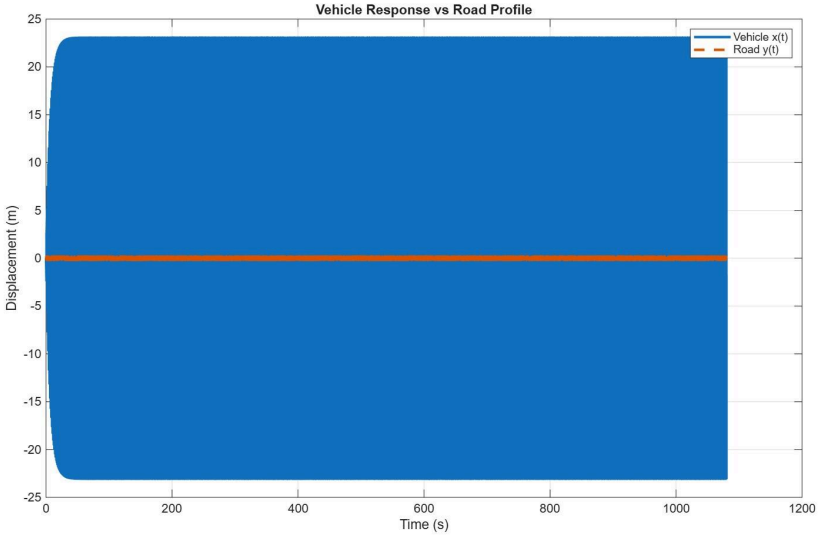


Figure B4. Resonance simulation over 6000 periods with increased scaling ($c \times 10 = 300$).

The initial amplitude rise is sharper and reaches lower peak values, stabilizing more quickly. The effect is due to the reduced overall displacement range.

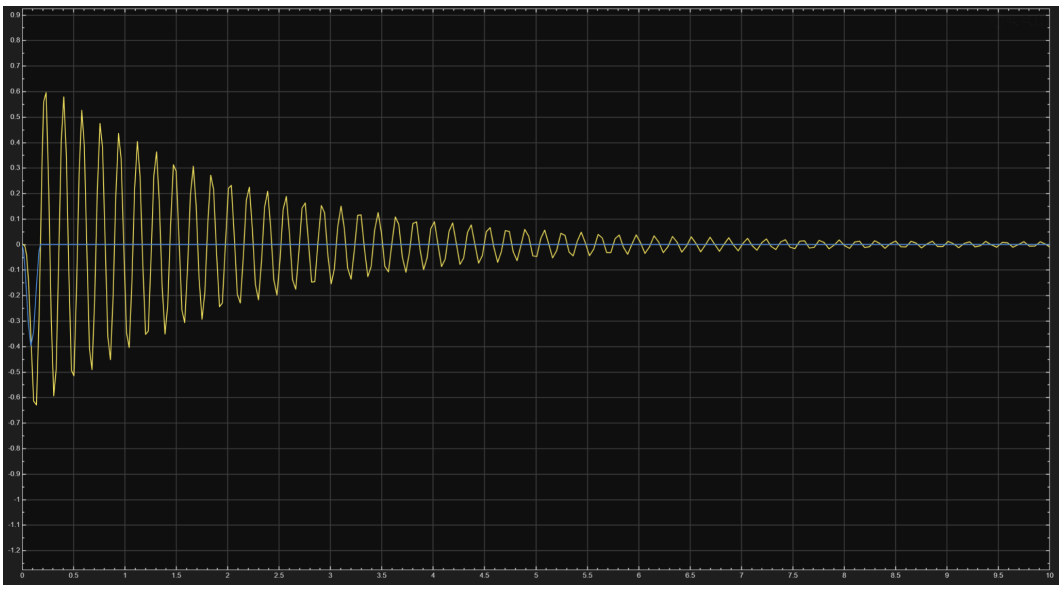


Figure C2. Displacement response for a single pothole with spring stiffness $k = 1218469.679\ \frac{N}{m}$;
($\omega_n = 34.9\ \frac{rad}{s} = \sqrt{1218469.679/1000}$).

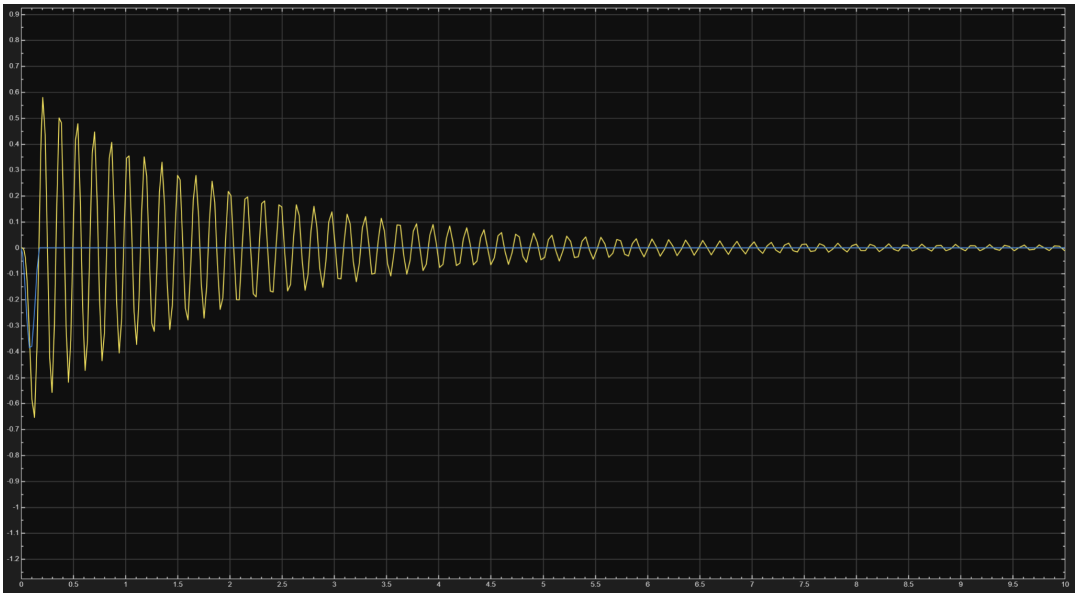


Figure C2. Displacement response for a single pothole with spring stiffness $k = 150000\ \frac{N}{m}$;
Just a bit more than resonance k value.

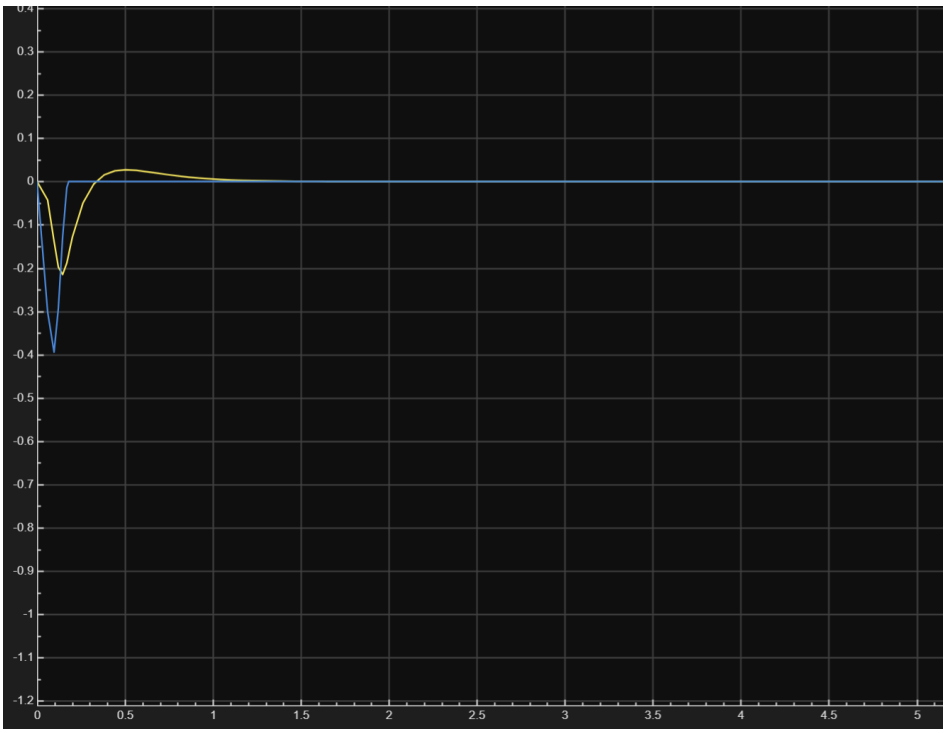


Figure C3. Displacement response for a single pothole with critical demping $c = 11832.16 \frac{kg}{s}$ and spring stiffness $k = 35000 \frac{N}{m}$.

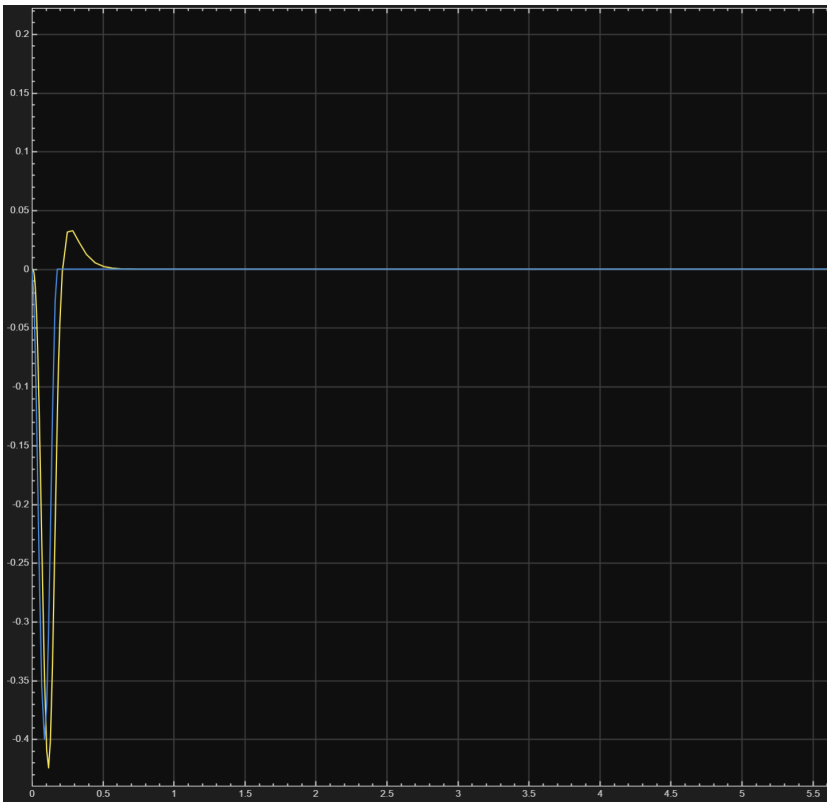


Figure C4. Displacement response for a single pothole with critical demping $c = 37416.57 \frac{kg}{s}$ and spring stiffness $k = 350000 \frac{N}{m}$.

mass_of_system.m × mass_of_system_int.m ×

C:\Users\Admin\Desktop\Classes\Dynamics\mass_of_system.m

```
1   function f=mass_of_system (t, x)
2
3   M=1000;
4   k=350*1000;
5   c=30;
6   V=(100*1000)/(60*60);
7   w=(2*pi)*V/5;
8   A=0.2;
9
10
11  Ft=A*sin(w*t);
12  Ft_der=A*w*cos(w*t);
13
14  % x(1) - x; x(2) - x_dot=V
15  f=[x(2);
16      (-c*(x(2)-Ft_der)-k*(x(1)-Ft))/M];
17  end
```

Figure D1. MATLAB code for this project: function defining the system of two first-order differential equations.

mass_of_system.m × mass_of_system_int.m ×

C:\Users\Admin\Desktop\Classes\Dynamics\mass_of_system_int.m

```
1   clear; clc; close all;
2
3   % Solve ODE
4   [t, x] = ode45(@mass_of_system, [0 0.54], [0 0]);
5
6   % Plot result
7   figure
8   plot(t, x(:,1), 'LineWidth', 2)
9   grid on
10  xlabel('Time (s)')
11  ylabel('Displacement x(t) (m)')
12  title('Vehicle Response x(t)')
```

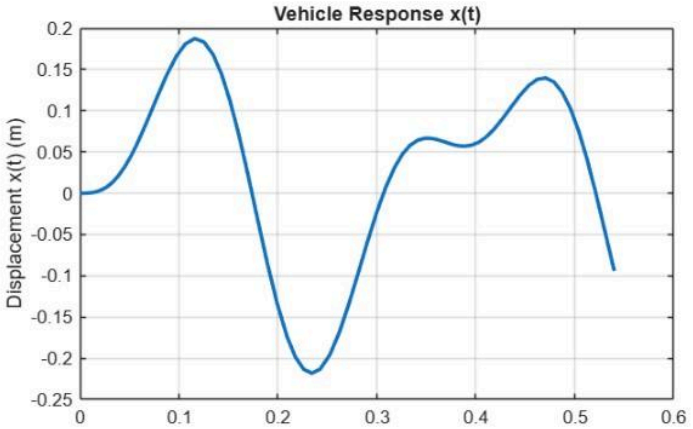


Figure D2. MATLAB code for this project: implementation using ode45 to solve the system and plot the displacement graph of the vehicle body.

```
13
14      % Example: find x at specific time
15      t_query = 0.1;
16      x_query = interp1(t, x(:,1), t_query);
17
18      disp(['x(', num2str(t_query), ') = ', num2str(x_query)])
```

```
x(0.1) = 0.17049
```

Figure D3. MATLAB code for this project: user-defined input to calculate vehicle displacement $x(t)$ at any time t; demonstrates an example of solving the differential equation via coding.

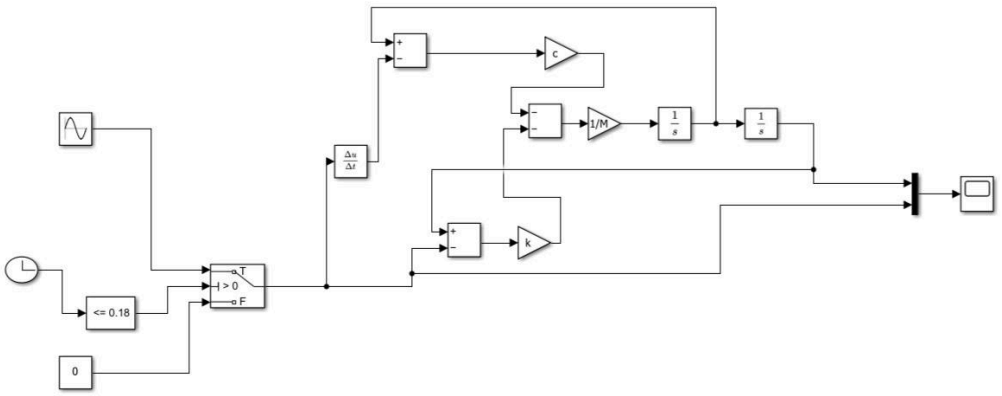


Figure D4. Simulink model of the vehicle suspension system for a single road pothole.